

Tautochrone Property of the Cycloid

Proof that the time to reach the lowest point of a cycloid is independent of starting position

1. Parametrization of the cycloid

We consider an inverted cycloid (brachistochrone curve) with parametric equations:

$$x = R(\theta - \sin \theta), \quad y = R(1 - \cos \theta), \quad 0 \leq \theta \leq \pi$$

where $\theta = 0$ corresponds to the top (cusp) and $\theta = \pi$ to the lowest point.

2. Arc length computation

$$\frac{dx}{d\theta} = R(1 - \cos \theta), \quad \frac{dy}{d\theta} = R \sin \theta$$

$$\begin{aligned} ds &= \sqrt{R^2(1 - \cos \theta)^2 + R^2 \sin^2 \theta} d\theta = R\sqrt{2 - 2 \cos \theta} d\theta \\ &= R\sqrt{4 \sin^2(\theta/2)} d\theta = 2R|\sin(\theta/2)| d\theta \end{aligned}$$

For $0 \leq \theta \leq \pi$, $\sin(\theta/2) \geq 0$, so:

$$ds = 2R \sin(\theta/2) d\theta$$

Arc length from top ($\theta = 0$) to θ :

$$s(\theta) = \int_0^\theta 2R \sin(\phi/2) d\phi = 4R [-\cos(\phi/2)]_0^\theta = 4R(1 - \cos(\theta/2))$$

3. Vertical coordinate

From the parametrization:

$$y = R(1 - \cos \theta) = 2R \sin^2(\theta/2)$$

4. Equation of motion

Let a particle start from rest at $\theta = \theta_0$ with arc length $s_0 = 4R(1 - \cos(\theta_0/2))$.
Energy conservation gives:

$$\frac{1}{2}mv^2 = mg(y - y_0) \Rightarrow v = \sqrt{2g(y - y_0)}$$

where $y_0 = 2R \sin^2(\theta_0/2)$.

Since $v = ds/dt$, we have:

$$\frac{ds}{dt} = \sqrt{2g(y - y_0)}$$

5. Time integral in terms of θ

Using $y = 2R \sin^2(\theta/2)$ and $ds = 2R \sin(\theta/2) d\theta$:

$$y - y_0 = 2R [\sin^2(\theta/2) - \sin^2(\theta_0/2)]$$

Thus:

$$v = \sqrt{4gR [\sin^2(\theta/2) - \sin^2(\theta_0/2)]}$$

The time from θ_0 to π is:

$$T = \int_{\theta_0}^{\pi} \frac{ds}{v} = \int_{\theta_0}^{\pi} \frac{2R \sin(\theta/2) d\theta}{\sqrt{4gR [\sin^2(\theta/2) - \sin^2(\theta_0/2)]}}$$

6. Change of variables

Let $u = \sin(\theta/2)$, $u_0 = \sin(\theta_0/2)$. Then:

$$du = \frac{1}{2} \cos(\theta/2) d\theta = \frac{1}{2} \sqrt{1 - u^2} d\theta \Rightarrow d\theta = \frac{2 du}{\sqrt{1 - u^2}}$$

Also:

$$ds = 2R \sin(\theta/2) d\theta = 2Ru \cdot \frac{2 du}{\sqrt{1 - u^2}} = \frac{4Ru du}{\sqrt{1 - u^2}}$$

The integrand becomes:

$$T = \int_{u=u_0}^1 \frac{4Ru du}{\sqrt{1 - u^2} \cdot \sqrt{4gR(u^2 - u_0^2)}} = \frac{2\sqrt{R}}{\sqrt{g}} \int_{u_0}^1 \frac{u du}{\sqrt{(1 - u^2)(u^2 - u_0^2)}}$$

7. Final integration

Let $u^2 = t$, $u du = dt/2$:

$$T = \frac{2\sqrt{R}}{\sqrt{g}} \int_{t=u_0^2}^1 \frac{dt/2}{\sqrt{(1 - t)(t - u_0^2)}} = \frac{\sqrt{R}}{\sqrt{g}} \int_{u_0^2}^1 \frac{dt}{\sqrt{(1 - t)(t - u_0^2)}}$$

This is a standard integral:

$$\int_a^b \frac{dt}{\sqrt{(b-t)(t-a)}} = \pi \quad \text{for } b > a$$

Here $a = u_0^2$, $b = 1$, so:

$$T = \frac{\sqrt{R}}{\sqrt{g}} \cdot \pi = \pi \sqrt{\frac{R}{g}}$$

8. Conclusion

The time $T = \pi\sqrt{R/g}$ is independent of u_0 , hence independent of the starting position θ_0 . Therefore, all balls released from rest at any point on the cycloid reach the lowest point in the same time.

$$\boxed{T = \pi \sqrt{\frac{R}{g}}}$$